K.S. INSTITUTE OF TECHNOLOGY, BANGALORE

Department of Computer Science & Design

DevOps Engineering Lab — BCSL657D — Lab Manual

Institution	K.S. Institute of Technology, Bangalore
Department	Computer Science & Design (CS&D)
Course	**DevOps Engineering Lab — BCSL657D**
Semester	VI Semester | Academic Year 2025–26
Program No.	**9 of 12**
Program Title	Introduction to Azure DevOps: Services Overview, Account Setup and Project Creation
Duration	~90 minutes (Account setup: 15 min | Explore services: 30 min | Repo init: 20 min | Viva: 15 min | Buffer: 10 min)
Pre-requisite	Microsoft account (Gmail/Outlook/college email) | Internet connection
Cost	ZERO — Azure DevOps is FREE. No credit card. No subscription required.
Azure DevOps URL	https://dev.azure.com
Organization	ksit-devops (dev.azure.com/ksit-devops)
Project	student-app-devops (Private, Git, main branch)

1. MOST IMPORTANT — No Credit Card Required

✓ Azure DevOps is completely FREE — no credit card, no billing, ever.

Azure DevOps (dev.azure.com) is a separate product from Azure Portal (portal.azure.com). Azure Portal → Cloud computing services → needs subscription and credit card Azure DevOps → DevOps tools (Repos, Pipelines, Boards) → FREE for all users The free tier of Azure DevOps includes: • Unlimited private projects • Unlimited Git repositories • 5 free users per organization • 1 free parallel pipeline job (Microsoft-hosted) • Azure Artifacts (2 GB free storage) Students should use: https://dev.azure.com — NOT portal.azure.com Any Microsoft account works: Gmail, Outlook, Hotmail, or college email.

⚠ Do NOT click 'Start Azure Free Trial' or enter credit card details:

If you accidentally land on portal.azure.com and see 'Start with an Azure free trial' — stop. That is the Azure cloud services portal — not what we need. The correct URL is always: https://dev.azure.com Bookmark it now. Programs 9, 10, and 11 all use this URL.

2. What This Program Is About

Programs 1–8 covered build automation (Maven, Gradle), CI/CD servers (Jenkins), and configuration management (Ansible) — all running locally on your machine. Programs 9–11 move to the cloud.

Azure DevOps is Microsoft's complete cloud-based DevOps platform. It provides everything a software team needs — code repositories, automated pipelines, project tracking, test management, and package hosting — all in one place, accessible from anywhere in the world.

Real-world connection:
Azure DevOps is used by thousands of companies globally — from small startups to Microsoft itself. When you join an IT company, there is a very high chance your team uses either Azure DevOps, GitHub Actions, or GitLab CI — all work on similar principles. Learning Azure DevOps now gives you a significant advantage in your first IT job. Companies like Infosys, Wipro, TCS, Cognizant, and Capgemini all use Azure DevOps for client projects. The skill you build in Programs 9–11 is directly applicable from day one of your career.

3. Learning Objectives

By the end of this lab, the student will be able to:

- Explain what Azure DevOps is and how it differs from Azure Portal
- Create an Azure DevOps account using a Microsoft account — without a credit card
- Create an Azure DevOps Organization and explain what an organization represents
- Create a Project within the organization with appropriate settings
- Identify and explain the purpose of all 5 Azure DevOps services
- Initialize a Git repository in Azure Repos and explain what initialization does
- Locate the repository clone URL and explain its structure
- Navigate Organization Settings and understand the free tier limits

4. Azure DevOps vs Azure Portal — Know the Difference

This is the most common confusion students face. Study this table before starting.

Aspect	Azure DevOps (dev.azure.com)	Azure Portal (portal.azure.com)
URL	https://dev.azure.com	https://portal.azure.com
Purpose	DevOps tools — Repos, Pipelines, Boards	Cloud infrastructure — VMs, databases, storage
Cost	FREE for 5 users — no credit card ever	Needs subscription — credit card required
What we use	Programs 9, 10, 11	NOT used in this course
Account needed	Any Microsoft account (Gmail works)	Azure subscription (paid or free trial)
When to use	Building and deploying software	Hosting servers, databases, cloud apps

5. The 5 Azure DevOps Services — What Each Does

Azure DevOps is not one tool — it is five integrated services. You will use Repos and Pipelines most in this course.

Service	URL Path	Purpose	Used in Lab
Boards	/_boards	Agile project management — work items, backlogs, sprints, kanban boards	Not used in Programs 9–11 — used for project planning in industry
Repos	/_git	Git-based source code repositories — store, branch, merge code	Program 9: Initialize repo. Program 10: Push student-app code here.
Pipelines	/_build	CI/CD automation — build, test, deploy code automatically on every commit	Program 10: Build pipeline. Program 11: Release pipeline.
Test Plans	/_testManagement	Manual and automated test management — test cases, runs, reports	Limited on free tier. Basic features available. Not used in these programs.
Artifacts	/_artifacts	Package hosting — publish and consume NuGet, Maven, npm packages	Artifacts feed 'ksit-devops' created automatically. Used in Program 10.

About Test Plans — free tier limitation:
When you click Test Plans, you see: 'Your current Azure DevOps license provides access to only a limited subset of Azure Test Plans features.' This is expected and normal on the free tier. Test Plans full features (complex test management, shared steps, configurations) require a paid license. For our lab, we only need Repos and Pipelines — both are fully free with no limitations.

6. Pre-Requisites — What Every Student Needs

✓	Requirement	Notes
☐	**A Microsoft account (Gmail, Outlook, Hotmail, or college email)**	*Students who already have Gmail can use it directly — no new account needed*
☐	**Internet connection**	*Azure DevOps is cloud-based — internet required throughout the lab*
☐	**Web browser (Chrome recommended)**	*Chrome works best with Azure DevOps UI*
☐	**NO credit card needed**	*Azure DevOps free tier — zero payment required at any step*
☐	**NO Azure subscription needed**	*dev.azure.com is separate from portal.azure.com*

7. Part A — Creating the Azure DevOps Account and Organization

A1. Navigate to Azure DevOps

1 **Open browser and go to: https://dev.azure.com**
Do NOT go to portal.azure.com — that is a different product

```
URL: https://dev.azure.com

  If already signed in to a Microsoft account:
  → You will see your profile page or an existing organization

  If not signed in:
  → Click 'Sign in' and use your Microsoft account
  → Gmail accounts work: sign in at login.microsoftonline.com
```

Which email to use:
Any of these work for Azure DevOps: • Gmail (e.g. yourname@gmail.com) — most students already have this • Outlook / Hotmail (e.g. yourname@outlook.com) • College email (e.g. yourrollno@ksit.edu.in) — recommended for academic use Teacher recommendation: Use your personal Gmail if you already use it for other Microsoft services. This ensures your Azure DevOps account remains accessible after college.

A2. Create a New Organization

2 **On the Azure DevOps profile page — click 'Create new organization'**
Organization = your company/college workspace in Azure DevOps

```
Organization name: ksit-devops
  → This becomes your URL: dev.azure.com/ksit-devops
  → Must be unique across all Azure DevOps users worldwide
  → If 'ksit-devops' is taken, use: ksit-devops-yourrollno

Region: India
  → Closest to Bangalore — best performance

Click: Continue
```

What is an Azure DevOps Organization?
An organization is the top-level container in Azure DevOps — like a company or college department. All projects, repositories, and pipelines belong to an organization. Structure: Organization: ksit-devops └── Project: student-app-devops ├── Repos (code) ├── Pipelines (CI/CD) └── Boards (work items) In a real company: Organization = Company name. Projects = individual product teams or client projects.

If you see 'You need an Azure subscription for billing':
This is a known screen that appears for some accounts. Do NOT click 'Get started with Azure' — that leads to billing. Simply click 'Continue' at the bottom right — this creates the organization without any billing. The subscription link is optional, not required for the free tier.

A3. Create the Project

3 **After organization is created — fill in the project details**
You will be taken directly to the 'Create a project to get started' screen

Field	Value to Enter
Project name	`student-app-devops`
Visibility	`Private (only you can see it)`
Version control	`Git (default — do not change)`
Work item process	`Basic or Agile (either is fine for this lab)`

4 **Click '+ Create project'**
Azure DevOps creates the project in seconds — no waiting

✓ **Project created successfully!**
You should now see the project dashboard at: dev.azure.com/ksit-devops/student-app-devops The left sidebar shows all 5 services: Overview, Boards, Repos, Pipelines, Test Plans, Artifacts. The project is Private — only you (and anyone you invite) can access it.

Observation A — Write your organization URL and project URL:

8. Part B — Exploring All 5 Azure DevOps Services

Click each service in the left sidebar and spend 2 minutes understanding what it contains. Do not create anything yet — just explore.

B1. Boards — Agile Project Management

5 **Click Boards in the left sidebar**
Boards manages work items — tasks, bugs, user stories

```
URL: dev.azure.com/ksit-devops/student-app-devops/_workitems/

  Sections visible under Boards:
  Work items  → Individual tasks or bugs
  Boards      → Kanban board (To Do / Doing / Done columns)
  Backlogs    → Product backlog — prioritized list of work
  Sprints     → Time-boxed iterations (2-week cycles typically)
  Queries     → Saved searches for work items
  Delivery Plans → Timeline view across multiple teams
```

Boards in a real company:

A Scrum master uses Sprints to plan 2-week work cycles. Developers pick tasks from the Backlog, move them through the Kanban board (To Do → In Progress → Done). Managers track progress in Delivery Plans. Azure Boards replaces physical sticky note boards used in Agile teams.

Observation B1 — List all sub-sections visible under Boards:

B2. Repos — Git Source Code Repository

6 **Click Repos in the left sidebar**
Repos hosts your Git repositories in the cloud

```
URL: dev.azure.com/ksit-devops/_git/student-app-devops

  When empty, Repos shows:
  → Clone URL (HTTPS): https://ksit-devops@dev.azure.com/ksit-
devops/student-app-devops
  → Option to push an existing repository from command line
  → Generate Git Credentials button

  Sections visible under Repos:
  Files       → Browse files and folders in the repository
  Commits     → History of all code changes
  Pushes      → List of push operations
  Branches    → All Git branches
  Tags        → Version tags
  Pull Requests → Code review requests
```

Azure Repos vs GitHub:
Both are cloud-based Git repository hosting services. Azure Repos is tightly integrated with Azure Pipelines (Programs 10 & 11). GitHub is more widely used for open-source projects. In industry: Azure Repos is preferred when the team uses Azure DevOps end-to-end. GitHub is preferred for open-source or when using GitHub Actions for CI/CD. Both use standard Git — the commands are identical.

Observation B2 — Write the HTTPS clone URL for your repository:

B3. Pipelines — CI/CD Automation

7 **Click Pipelines in the left sidebar**
Pipelines is where CI/CD magic happens — Programs 10 and 11 live here

```
URL: dev.azure.com/ksit-devops/student-app-devops/_build

  Currently shows: 'Create your first Pipeline'
```

```
This is what Programs 10 and 11 will configure.

Sections under Pipelines:
Pipelines    → CI build pipelines (Program 10)
Environments → Deployment targets (Program 11)
Library      → Shared variables and secure files
```

What Pipelines replaces in our course:

Programs 5 and 6 used Jenkins (running locally) for CI. Program 10 uses Azure Pipelines (running in the cloud) for CI. The concept is identical — on every code commit, the pipeline builds and tests your code. The difference is Jenkins runs on your machine; Azure Pipelines runs on Microsoft's cloud servers (called Microsoft-hosted agents). You do not need to install or maintain anything.

Observation B3 — Write what text appears on the empty Pipelines page:

B4. Test Plans — Test Management

8 **Click Test Plans in the left sidebar**
Note the free tier limitation message

```
URL: dev.azure.com/ksit-devops/student-app-devops/_testManagement/

  You will see:
  'Unlock the full power of Azure Test Plans'
  'Your current Azure DevOps license provides access
   to only a limited subset of Azure Test Plans features.'

  This is expected on the free tier.
  Basic test execution features are available.
  Advanced features need a paid Test Plans license.

  For Programs 9-11: Test Plans is NOT used.
  Our tests run automatically inside Pipelines (Program 10).
```

Observation B4 — Write the exact message shown on the Test Plans page:

B5. Artifacts — Package Repository

9 **Click Artifacts in the left sidebar**
Artifacts stores compiled packages — JAR files, npm modules, NuGet packages

```
URL: dev.azure.com/ksit-devops/student-app-devops/_artifacts/

  A feed named 'ksit-devops' was created automatically
  when you created the organization.
```

```
What Artifacts does:
→ Stores Maven JARs (like student-app-1.0-SNAPSHOT.jar)
→ Stores npm packages (Node.js)
→ Stores NuGet packages (.NET)
→ Acts as a private Maven Central for your organization

Free tier: 2 GB storage
```

Artifacts vs local .m2 cache:
In Programs 1–8, Maven stored downloaded JARs in C:\Users\lenovo\.m2\repository — the local cache. Azure Artifacts is the cloud equivalent — a shared repository that your entire team can publish to and download from. In a company: your Jenkins or Azure Pipeline publishes the built JAR to Artifacts. Other developers or downstream pipelines can then download this JAR as a dependency. No more emailing JAR files or copying them manually.

Observation B5 — Write the feed name shown in Artifacts:

9. Part C — Initializing the Repository

An empty repository is not useful. Initializing adds a README.md file and creates the first commit — this sets up the main branch and makes the repository ready for code to be pushed.

C1. Initialize the Repository with a README

10 **Click Repos → Click Initialize**
This creates the first commit and the main branch

```
Navigation: Repos → Files

  If repository is empty, scroll down to find:
  'Initialize' button OR
  'Initialize with a README or gitignore' section

  Click Initialize (or Initialize repository)

  Azure DevOps will:
  1. Create README.md with template content
  2. Create the 'main' branch
  3. Make the first commit: 'Added README.md'
  4. Show the Files view with README.md visible
```

✓ Repository initialized!
You should now see: • README.md file in the file browser • Branch: main (shown at the top) • Commit hash: be02673f (your hash will be different) • Commit message: Added READ... • 'Set up build' button visible — this leads to Pipeline creation (Program 10)

C2. Explore the Repository Structure

11 **Note the repository URL structure and clone URL**
This URL will be used in Program 10 to push code

```
Clone URL (HTTPS):
https://ksit-devops@dev.azure.com/ksit-devops/student-app-devops

URL structure breakdown:
  https://                          → Protocol
  ksit-devops@                      → Organization name (authentication
hint)
  dev.azure.com/                    → Azure DevOps domain
  ksit-devops/                      → Organization name
  student-app-devops                → Project name = Repository name

To push code to this repo (used in Program 10):
  git remote add origin https://ksit-devops@dev.azure.com/ksit-
devops/student-app-devops
  git push -u origin main
```

Observation C1 — Write your full HTTPS clone URL:

10. Part D — Organization Settings and Free Tier Limits

12 **Click Organization Settings at the bottom left of the sidebar**
This shows billing, users, and parallel job limits

```
Navigation: Bottom left → Organization settings
URL: dev.azure.com/ksit-devops/_settings/

  Key sections to explore:
  Overview     → Organization name, creation date
  Users        → Manage team members (5 free on free tier)
  Billing      → Current plan (Free tier)
  Parallel jobs → How many pipelines can run simultaneously

  Under Parallel jobs (Pipelines section):
  Microsoft-hosted: 1 free parallel job
  Self-hosted: Unlimited free parallel jobs
```

Feature	Free Tier Limit	What this means
Users per organization	5 free users	Up to 5 people can collaborate — perfect for a lab group
Private projects	Unlimited	Create as many projects as needed — all private and free
Git repositories	Unlimited	Each project can have multiple repositories

Feature	Free Tier Limit	What this means
Microsoft-hosted parallel jobs	1 job	One pipeline can run at a time on cloud agents
Self-hosted parallel jobs	Unlimited	Run pipelines on your own machine — no limit
Azure Artifacts storage	2 GB	Store built packages — more than enough for lab work
Test Plans	Limited subset	Basic test execution free — advanced features need license

📋 Observation D1 — Write the number of Microsoft-hosted parallel jobs on your free tier:

11. Program 9 Summary — What Was Accomplished

#	Action	Result	Used in
1	Created Azure DevOps account using raghav.ksit@gmail.com	Microsoft account linked to Azure DevOps — free, no card	Programs 9, 10, 11
2	Created Organization: ksit-devops	URL: dev.azure.com/ksit-devops — India region	Programs 9, 10, 11
3	Created Project: student-app-devops	Private project with Git version control	Programs 9, 10, 11
4	Explored all 5 services	Boards, Repos, Pipelines, Test Plans, Artifacts	Understanding only
5	Initialized repository	README.md added, main branch created, first commit: be02673f	Programs 10, 11
6	Noted clone URL	https://ksit-devops@dev.azure.com/ksit-devops/student-app-devops	Program 10
7	Explored Organization Settings	Free tier: 5 users, 1 parallel job, unlimited repos	Understanding

12. Expected Outputs — What to Record in Lab Journal

#	Screenshot	What to Show	Marks
1	Azure DevOps profile page	Raghav S, raghav.ksiit@gmail.com, Microsoft account	Account created
2	Organization created	URL showing ksit-devops, 'Create a project' screen	Organization setup

#	Screenshot	What to Show	Marks
3	Project dashboard	student-app-devops with all 5 services in sidebar	Project created
4	Boards — Work items	Empty work items page with sub-sections visible	Service explored
5	Repos — Empty	Clone URL visible: https://ksit-devops@dev.azure.com/...	Repo URL noted
6	Pipelines — Empty	'Create your first Pipeline' screen visible	Service explored
7	Test Plans — Limited	Free tier limitation message visible	Service explored
8	Artifacts — Feed	'ksit-devops' feed visible, 'Connect to feed' screen	Service explored
9	Repos — Initialized	README.md visible, main branch, first commit hash	Repo initialized
10	Organization Settings	Free tier info — 5 users, parallel jobs	Settings explored

13. Common Issues and Fixes

Issue	Cause	Fix
Seeing 'Start Azure free trial' and credit card request	Accidentally opened portal.azure.com	Close tab. Open new tab. Go to https://dev.azure.com — completely different site
Organization name 'ksit-devops' is already taken	Someone else registered this name globally	Use: ksit-devops-2025 or ksit-devops-yourrollno (must be globally unique)
'You need an Azure subscription for billing' message	Azure checking for subscription link	Click 'Continue' button — subscription is optional, not required for free tier
Cannot find 'Initialize' button in Repos	Repository page shows clone URL instead	Scroll down on the Repos/Files page — Initialize button is below the clone URL section
Test Plans shows upgrade message	Free tier limitation	This is normal and expected. Test Plans advanced features need a paid license. Ignore for this lab.
Forgot account credentials	Lost Microsoft account password	Go to account.microsoft.com → Reset password → Then sign in to dev.azure.com again

14. Viva-Voce Question Bank

Level 1 — Basic (Every student must answer these)

#	Question	Expected Answer
1	**What is Azure DevOps and what is the URL to access it?**	Azure DevOps is Microsoft's cloud-based DevOps platform providing Repos, Pipelines, Boards, Test Plans, and Artifacts. URL: https://dev.azure.com. It is free for up to 5 users — no credit card required.
2	**What is the difference between Azure DevOps and Azure Portal?**	Azure DevOps (dev.azure.com) provides DevOps tools — free, no credit card. Azure Portal (portal.azure.com) provides cloud infrastructure (VMs, databases) — needs a paid subscription. We use only Azure DevOps in this course.
3	**Name all 5 Azure DevOps services and what each does.**	Boards (Agile project management — work items, sprints), Repos (Git source code hosting), Pipelines (CI/CD automation), Test Plans (test management — limited on free tier), Artifacts (package hosting — JARs, npm, NuGet).
4	**What did you create in this lab? Give the full URL of your project.**	Organization: ksit-devops. Project: student-app-devops. Full URL: dev.azure.com/ksit-devops/student-app-devops. Repository initialized with README.md on main branch.
5	**Why did we initialize the repository with a README? What did that create?**	Initializing creates the first commit, establishes the main branch, and adds README.md. Without initialization, the repository is empty — no branch exists and Git operations like push/pull cannot work until at least one commit is present.

Level 2 — Conceptual

#	Question	Expected Answer
1	**What is an Azure DevOps Organization? How is it different from a Project?**	Organization is the top-level container — like a company or college. It has one URL and billing plan. Projects are inside organizations — each project has its own Repos, Pipelines, and Boards. One organization can have many projects. Structure: Organization (ksit-devops) → Project (student-app-devops) → Services.
2	**The free tier allows 1 Microsoft-hosted parallel job. What does this mean in practice?**	Only 1 pipeline can run at a time on Microsoft's cloud build agents. If 2 developers push code simultaneously, the second pipeline waits in a queue. For a lab with one student this is not an issue. For a team, additional parallel jobs require payment.

#	Question	Expected Answer
3	**How is Azure Repos similar to what we used in Programs 1–8? What is the key difference?**	Azure Repos is a cloud Git repository — same as the .git folder and commits concept. In Programs 1–8, code was only on the local machine (D:\Exercise\student-app). Azure Repos stores it in Microsoft's cloud — accessible from any device, anywhere, with full history and collaboration features.
4	**A company has 50 developers. They want to use Azure DevOps free tier. What is the limitation?**	The free tier only supports 5 users per organization for full access. For 50 developers, additional Basic licenses are needed (approx $6 per user per month). The 5 free users get full access to all services except Test Plans which needs a separate license.
5	**Why is 'Private' visibility recommended for student projects in Azure DevOps?**	Private means only people you explicitly invite can see the project. Public projects are visible to everyone on the internet — including your code, build history, and pipeline configurations. For learning projects containing institutional information or personal credentials, Private is always safer.

15. Quick Reference Card

Item	Detail
`Azure DevOps URL`	https://dev.azure.com
`Organization`	ksit-devops
`Organization URL`	https://dev.azure.com/ksit-devops
`Project`	student-app-devops
`Project URL`	https://dev.azure.com/ksit-devops/student-app-devops
`Repos URL`	https://dev.azure.com/ksit-devops/_git/student-app-devops
`Clone URL (HTTPS)`	https://ksit-devops@dev.azure.com/ksit-devops/student-app-devops
`Pipelines URL`	https://dev.azure.com/ksit-devops/student-app-devops/_build
`Boards URL`	https://dev.azure.com/ksit-devops/student-app-devops/_boards
`Org Settings URL`	https://dev.azure.com/ksit-devops/_settings/
`Account email`	raghav.ksit@gmail.com
`Cost`	ZERO — no credit card — no subscription
`Free users`	5 per organization
`Free parallel jobs`	1 Microsoft-hosted
`Repository initialized`	README.md, main branch, commit be02673f

— End of Program 9 Lab Manual —